Seattle and Baltimore Rain Comparison

I. Introduction

This report aims to determine if it rains more in Seattle than Baltimore, MD. There are a couple of different approaches to determine if it rains more in one city versus another. 'More' can refer to either the measurement of rainfall or the frequency of days when it rains. However, conceptualizing a given numerical amount of rainfall can be difficult, which makes the measurement of rainfall slightly less applicable. However, the amount of rainfall can give some indication to the duration of rain. If we assume, in general, the larger the amount of rainfall, the longer the rain event could have occurred. Therefore, it is also important to take into account the number of rainy days or proportion of days with rain. This report aims to compare the rainfall in Baltimore, MD and Seattle, WA by comparing the amount of rainfall and proportion of days with rain over a 5-year period, January 1st, 2018, to December 31st, 2022.

II. Methods

First and foremost, this analysis uses data from NOAA National Centers for Environmental Information. The Climate Data Online provides access to an archive of historical weather and climate data for a specified weather station. NOAA provides several various weather-related observations; however, this analysis will focus on precipitation, measured in inches. Any missing dates were added with null values for precipitation. To approximate missing precipitation values, each null precipitation observation was replaced with the average amount of precipitation for its corresponding day of the year in that city. Subsequently, to compare the amount of precipitation, the mean average daily precipitation and 5-number summary were calculated for the two cities. A t-test was performed to determine the statistical significance of the difference in average monthly precipitation. To compare the frequency of rainy days, the ratio of days with precipitation was calculated. A z-test was performed to determine the statistical significancy of the difference in proportion of days with rain for the two cities by month.

III. Results

The analysis determined that Seattle has an overall lower average amount of precipitation. Seattle has an average daily precipitation of 0.113 inches with a standard deviation of 0.241 inches. Baltimore has an average daily precipitation of 0.142 inches with a standard deviation of 0.377 inches. Baltimore has a considerably higher maximum precipitation value, 4.79 inches compared to 2.60 inches. Figure 1 shows the difference in average daily precipitation values between the two cities. Figure 2 shows the average precipitation per month for each city. The statistically significant differences are indicated with an asterix. The results of the t-test are as follows:

Month	Baltimore Average	Seattle Average	t-statistic	p-value
1	0.145876	0.230742	2.24	0.026
2	0.144356	0.176472	0.71	0.476
3	0.123016	0.089075	-1.11	0.268

4	0.15175	0.100483	-1.34	0.181
5	0.102968	0.069161	-1.37	0.173
6	0.08865	0.063167	-1.13	0.258
7	0.126887	0.013984	-5.05	0.000
8	0.153995	0.019995	-4.98	0.000
9	0.214917	0.055622	-3.06	0.003
10	0.128065	0.118452	-0.3	0.763
11	0.155667	0.201867	1.26	0.210
12	0.164903	0.224903	1.26	0.210

Seattle appears to have a higher precipitation amount in the winter months, comparable amounts of precipitation in the spring and fall, lower amounts of precipitation in the summer. However, the months with a statistically significant difference are January, July, August, and September; therefore, the most significant differences are in the late summer.

When comparing the proportion of rainy days overall, Seattle has a higher proportion of days with precipitation than Baltimore. Figure 3 shows the proportion of days with precipitation for each city. To provide a more granular analysis, Figure 2 shows the proportion of rainy days per month for the two cities, with an asterisk for statistically significant differences. The results of the z-test are as follows:

Month	Baltimore Proportion	Seattle Proportion	z-statistic	p-value
1	0.283871	0.567742	-5.05	0.000
2	0.205674	0.496454	-5.12	0.000
3	0.258065	0.335484	-1.49	0.136
4	0.346667	0.553333	-3.60	0.000
5	0.322581	0.464516	-2.56	0.011
6	0.340000	0.406667	-1.19	0.233
7	0.380645	0.200000	3.50	0.000
8	0.348387	0.212903	2.65	0.008
9	0.326667	0.320000	0.12	0.902
10	0.393548	0.541935	-2.62	0.009
11	0.360000	0.606667	-4.27	0.000
12	0.374194	0.825806	-8.12	0.000

Most months had a statistically significant difference in proportion, except for March, June, and September. Notably September has a statistically significant difference in the amount of rain but not overall proportion of days with rain.

IV. Conclusion

Interestingly, this analysis has demonstrated a complicated conclusion. It is difficult to definitively announce which city has more rain. If we compare overall average rainfall, the answer is Baltimore; if we compare the proportion of days with rainfall, the answer is Seattle. In terms of average amount of rainfall, Baltimore has a higher average daily amount of rainfall and a higher maximum rainfall. Baltimore often tends to have a more consistent monthly average rainfall than Seattle, which exhibits seasonal changes in precipitation. When there is a rainy day in Baltimore, it tends to result in a higher amount of rainfall than the average rainy day in Seattle, despite the latter city having a higher frequency of rainy days.

It is important to note that this report was limited by the lack of timestamps for weather observations. Which makes it difficult to estimate the intensity and duration of rainfall events. For example, a daily total of 1 inch of rainfall over a 10-hour period may indicate a light rain event, despite the long duration. 1 inch of rainfall over 2 hours would indicate a much more intense rain event and be more likely to have a bigger impact on the city. Additionally, future studies should include some threshold for extreme rain events that have a significant impact on the city. Severe rain leading to mudslides, reduced visibility, or increased traffic accidents can help contextualize the amount of rain received. Understanding these additional variables would result in a more extensive understanding of the rain in each city.

V. Appendix

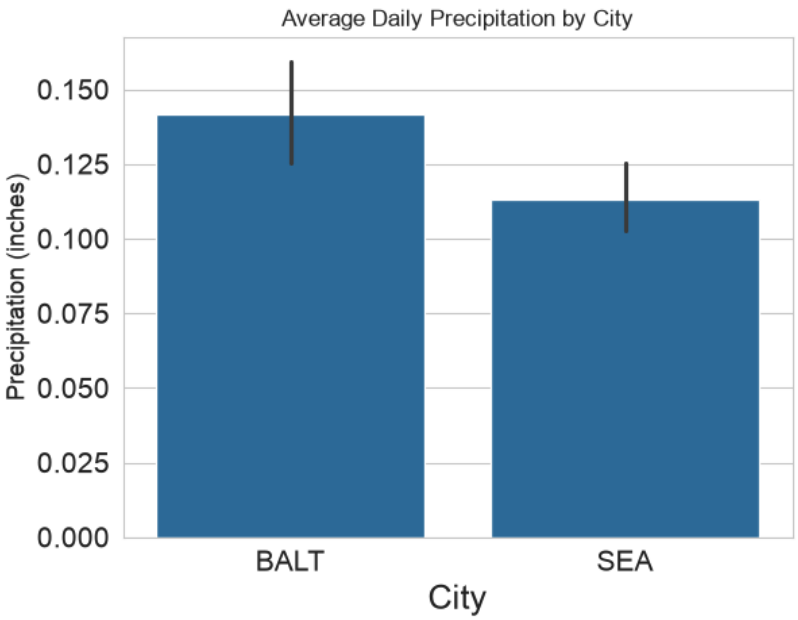


Figure 1: 'Average Daily Precipitation by City.' Shows the average daily precipitation amounts in inches by city, including the confidence intervals. Baltimore is on the left, and Seattle is on the right.

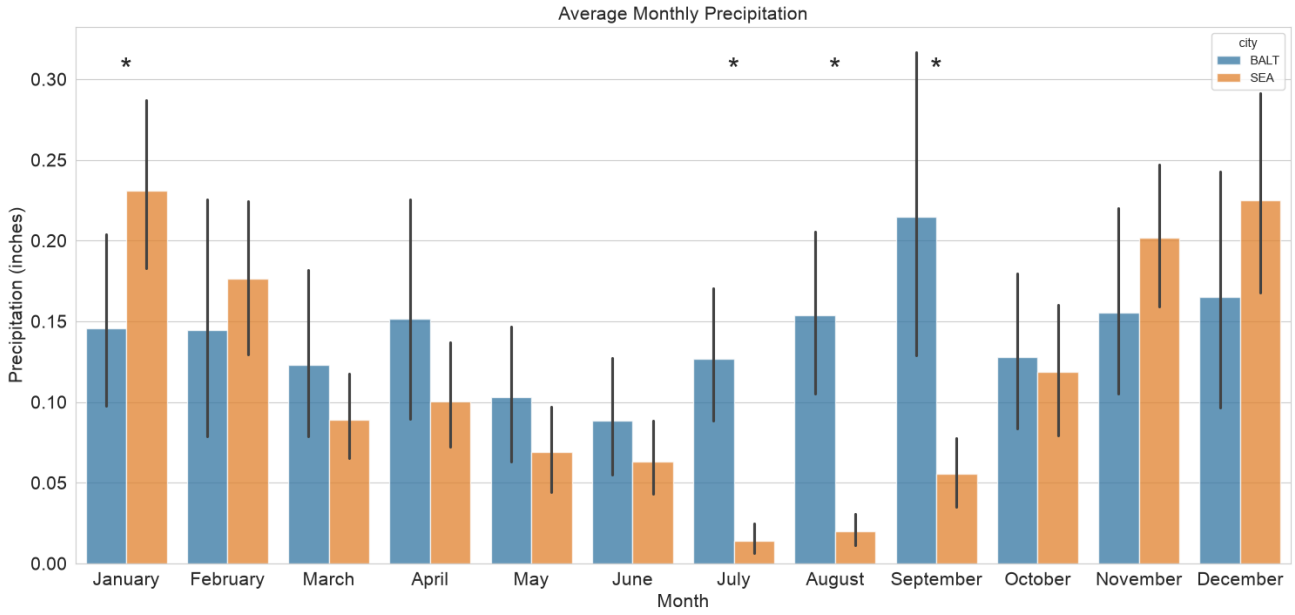


Figure 3: "Average Monthly Precipitation." This graph shows the average precipitation across 5 years by month and city. It includes confidence intervals and asterixis above months determined to have a statistically significant difference in average precipitation between the Baltimore, blue, and Seattle, orange.

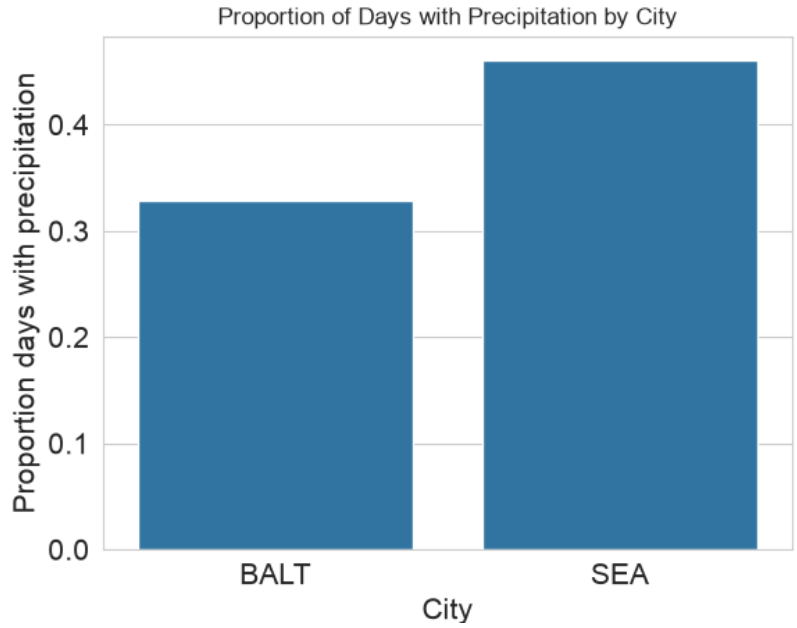


Figure 4: 'Proportion of Days with Precipitation by City.' The bar graph shows the proportion of days with precipitation of Baltimore (left) and Seattle (right).

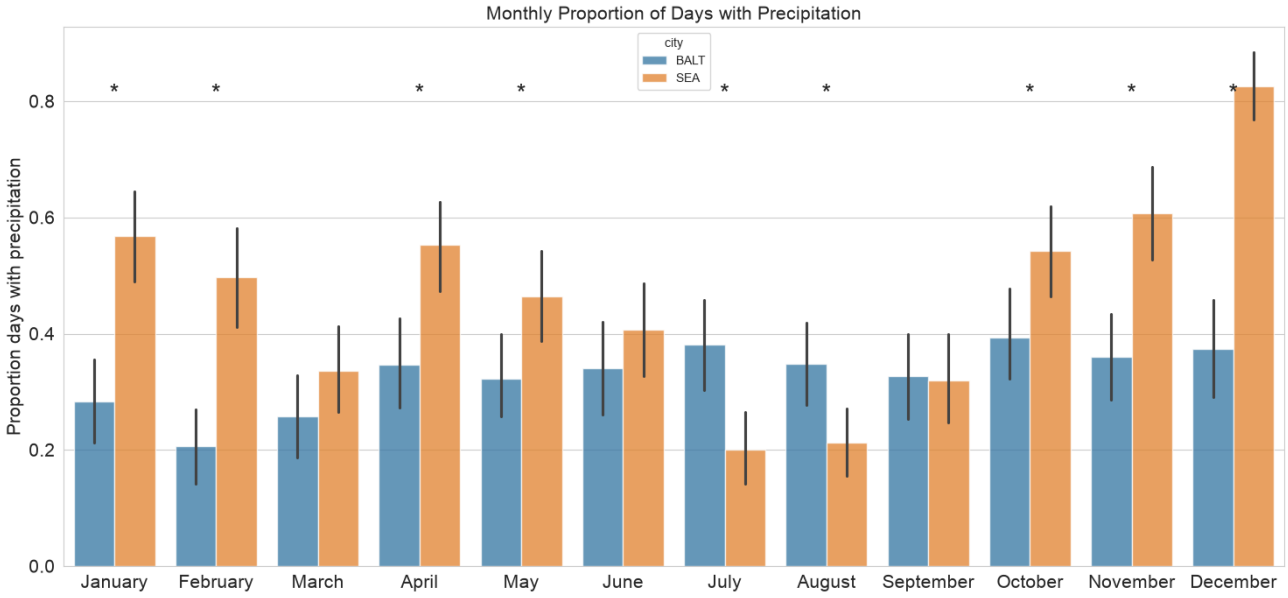


Figure 5: "Monthly Proportion of Days with Precipitation." This graph plots the proportion of days with precipitation per month for each city, with Baltimore in blue and Seattle in orange. It includes confidence intervals and asterisks above months determined to have a statistically significant difference in proportion.